

SULPHUR SPRINGS, TX, USA

WMO: 722362

Lat: 33.167N

Lon: 95.617W

Elev: 489

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 93937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.4	28.0	11.1	9.9	31.3	15.7	12.4	34.9	27.6	49.7	25.4	48.7	9.1	0	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	19.6	99.9	73.3	98.2	73.4	95.5	73.8	78.1	89.2	77.3	88.2	76.7	87.4	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	135.5	81.3	74.7	132.5	81.1	73.7	127.9	80.8	41.8	88.8	41.0	88.1	40.3	87.5	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.0	19.7	17.6	17.7	102.8	3.1	2.5	15.4	104.6	13.6	106.0	11.9	107.4	9.6	109.2		
			16.0	79.2	3.0	1.2	13.8	80.1	12.1	80.8	10.4	81.5	8.2	82.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.1	46.0	49.8	58.1	64.8	72.9	81.5	84.2	84.8	78.2	66.9	56.8	48.3
	DBStd	15.73	9.72	10.42	9.40	7.49	6.47	3.92	4.16	4.66	5.81	8.35	9.30	9.47
	HDD50	546	192	122	36	4	0	0	0	0	0	2	40	150
	HDD65	2261	591	433	249	95	16	0	0	0	78	275	524	
	CDD50	6423	68	116	287	447	710	944	1060	1078	847	527	243	96
	CDD65	2666	2	8	35	88	261	494	595	613	398	138	28	6
	CDH74	29688	11	34	194	635	2259	5647	7464	7839	4192	1245	143	25
	CDH80	14277	0	2	17	108	757	2698	3983	4330	1978	387	17	0
Wind	WSAvg	7.6	8.3	8.8	9.1	9.1	7.8	7.4	6.7	6.0	5.9	7.0	7.9	7.9
Precipitation	PrecAvg	45.9	3.3	3.6	4.3	3.8	4.6	4.1	3.1	2.4	3.4	4.8	4.1	4.0
	PrecMax	67.1	7.2	8.7	10.2	7.9	11.4	14.5	8.8	5.8	8.9	15.7	13.0	8.3
	PrecMin	22.4	0.0	0.5	0.5	1.3	1.3	0.6	0.1	0.1	0.8	0.3	0.7	0.4
	PrecStd	11.4	1.8	2.2	2.3	1.8	2.6	3.1	2.7	1.5	1.9	3.7	3.0	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.3	77.6	82.1	86.8	92.3	98.9	102.1	104.2	99.3	91.2	82.2	77.2
		MCWB	59.0	63.4	64.1	68.0	72.5	73.0	73.4	72.6	70.5	70.8	67.3	64.0
	2%	DB	71.6	73.7	79.1	82.4	89.9	96.0	99.4	100.4	95.9	87.9	78.2	72.7
		MCWB	60.2	62.0	64.3	67.9	72.1	73.2	73.1	73.8	72.5	71.2	65.3	63.0
	5%	DB	67.1	70.6	75.6	80.8	87.3	93.4	97.2	99.0	93.1	84.5	74.6	69.0
		MCWB	58.1	61.3	62.5	66.7	71.6	73.7	73.9	73.9	72.5	68.4	64.2	60.9
	10%	DB	63.1	67.3	72.7	77.9	84.3	91.2	94.8	96.5	90.6	81.3	72.0	64.3
		MCWB	54.4	59.3	61.9	65.1	70.7	74.0	74.5	73.9	71.6	66.7	63.3	57.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.8	67.9	68.8	73.3	76.3	79.0	78.9	79.2	77.6	75.5	71.3	68.4
		MCDB	71.0	72.9	76.9	81.0	84.9	88.9	90.4	91.8	87.2	83.7	78.6	72.8
	2%	WB	63.0	66.0	67.2	70.9	74.7	77.6	77.9	78.1	76.4	74.1	69.0	65.7
		MCDB	67.7	71.6	74.5	79.3	83.9	87.1	89.1	90.1	86.1	82.3	74.2	71.0
	5%	WB	60.0	63.6	65.7	69.3	73.5	76.6	77.2	77.3	75.4	72.8	67.0	62.6
		MCDB	65.3	68.1	72.5	77.0	83.1	86.4	88.3	88.9	84.9	80.6	72.7	67.2
	10%	WB	55.3	60.2	64.0	67.9	72.5	75.7	76.4	76.5	74.6	70.9	64.5	59.0
		MCDB	61.6	66.2	70.8	75.3	81.6	85.8	87.5	87.8	84.2	78.4	70.6	63.9
Mean Daily Temperature Range	5% DB	MDBR	18.6	18.2	19.2	19.7	18.3	18.2	18.8	19.6	20.3	21.0	19.9	18.3
		MCDBR	24.6	22.2	21.4	21.1	20.0	20.3	21.2	22.4	22.7	22.4	21.0	24.2
		MCWBR	15.7	14.2	11.1	10.0	7.3	5.3	4.6	4.8	6.5	8.4	11.4	16.2
	5% WB	MCDBR	21.2	19.4	17.7	17.6	17.8	17.2	17.9	18.7	18.4	18.1	17.7	20.9
		MCWBR	17.0	14.5	11.4	10.8	7.7	6.0	5.4	5.1	6.0	9.1	13.7	16.6
Clear-Sky Solar Irradiance	taub	0.316	0.331	0.358	0.392	0.427	0.439	0.450	0.448	0.401	0.357	0.340	0.320	
	taud	2.489	2.460	2.404	2.320	2.283	2.288	2.288	2.294	2.286	2.372	2.472	2.467	2.486
	Ebn at Noon	286	292	292	285	274	270	266	265	274	279	274	278	
	Edh at Noon	27	31	36	41	43	42	42	42	36	30	28	26	
All-Sky Solar Radiation	RadAvg	866	993	1361	1673	1818	2027	2061	1911	1601	1265	928	762	
	RadStd	82	152	113	115	158	197	134	137	148	182	118	99	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page